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MULTIMEDIA DATABASE

Cross-Modal Implicit Relation Reasoning and Aligning for Text-to-Image Person Retrieval

IRRA • CVPR 2023

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Presented by: [FULL NAME – STUDENT ID – GROUP]

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ONE LOVE. ONE FUTURE.

I. Introduction — From Description to Top-k

TEXT QUERY

**“A man
wearing a
white-and-gray
striped shirt,
green pants,
and shoes.”**

INPUT

Natural-language
description

OUTPUT

Ranked image
list

Baseline



IRRA

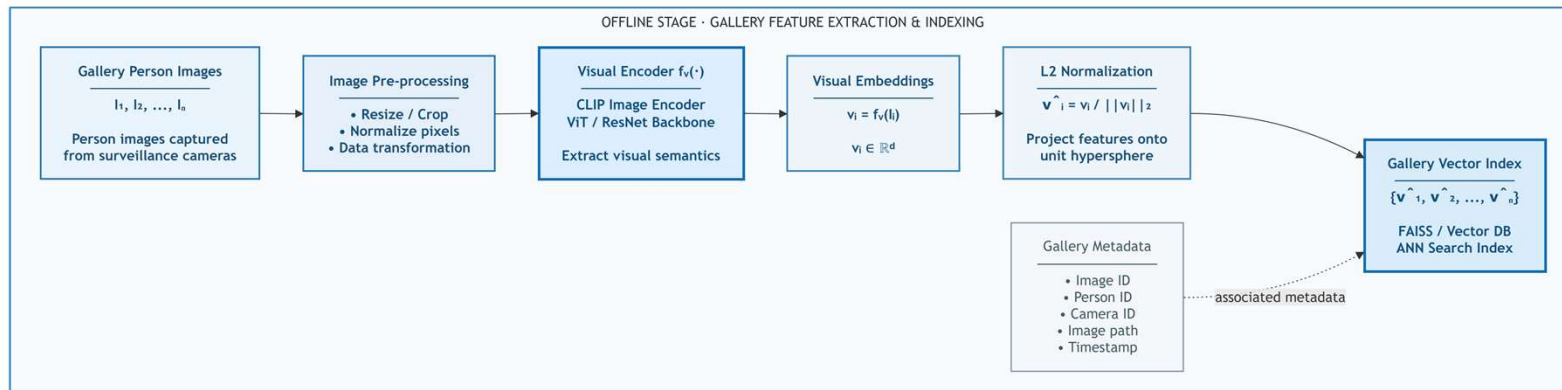


Cross-modal retrieval

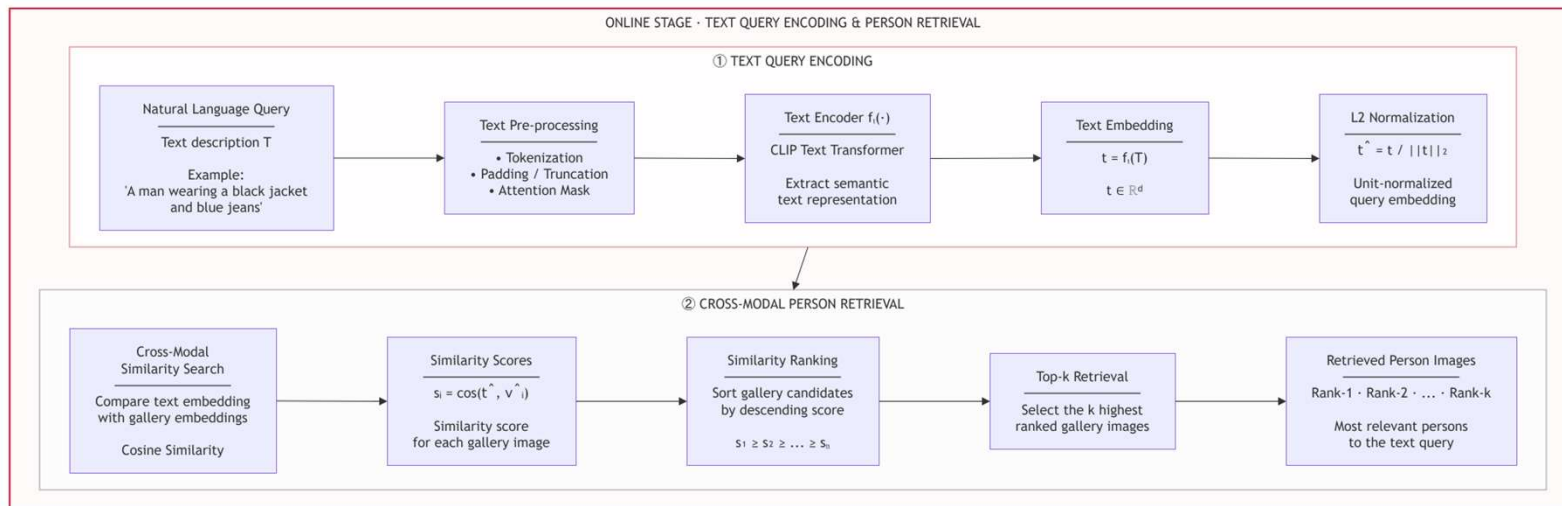
query = text → gallery = images → retrieve Top-k

I. Introduction — Multimedia Database Perspective

OFFLINE • GALLERY INDEXING



ONLINE • QUERY PROCESSING



I. Introduction — Why Is the Task Difficult?

1

Large intra-identity variation

Viewpoint • pose • lighting • occlusion

2

Modality heterogeneity

Pixels/patches ↔ tokens and word order

3

Subtle discriminative details

White/black bag • striped/plain shirt • shoes

4

Language ambiguity

Arbitrary order • missing attributes • synonyms

Learn fine-grained local cues while retaining efficient global retrieval.

I. Background — Three Matching Paradigms

GLOBAL

One image vector $\leftrightarrow$ one text vector

Fast, but misses fine details

EXPLICIT LOCAL

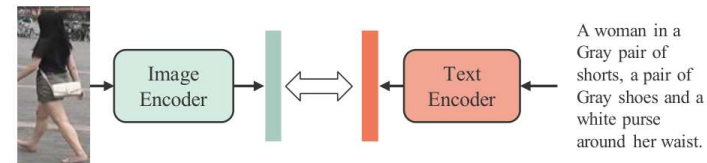
Body parts $\leftrightarrow$ textual phrases

Noisy priors; multiple vectors to store and compare

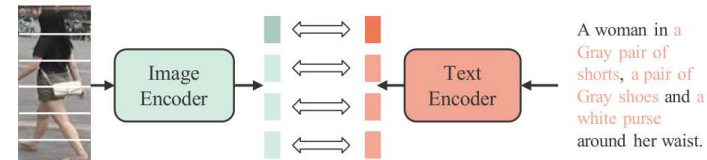
IRRA

Implicit local learning during training

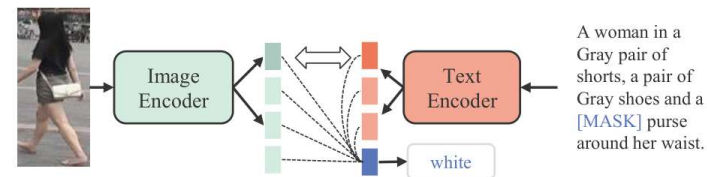
Global embeddings at inference



(a) Early global matching paradigm



(b) Existing explicit local matching paradigm



(c) Our implicit relation reasoning aided matching paradigm

$\longleftrightarrow$ Alignment ■ Global Image Feature ■ Global Text Feature
..... Attention ■ Local Image Feature ■ Local Text Feature ■ Masked Token

I. Background — Dual-Stream Models and CLIP

SINGLE-STREAM

[image tokens + text tokens]



JOINT TRANSFORMER

- + Strong cross-modal interaction
- Recomputed for every image–text pair

DUAL-STREAM • CLIP

IMAGE
ENCODER

TEXT
ENCODER

joint embedding
space

- + Precomputed gallery embeddings
- + Image–text pre-training
- Limited fine-grained interaction

IRRA = full CLIP + interaction only during TRAINING

II. Proposed Method — Research Question

How can fine-grained image–word relations improve global embeddings
without part-level supervision or additional retrieval cost?

1

IRR / MLM

Implicitly learns fine-grained image–text relations

2

SDM

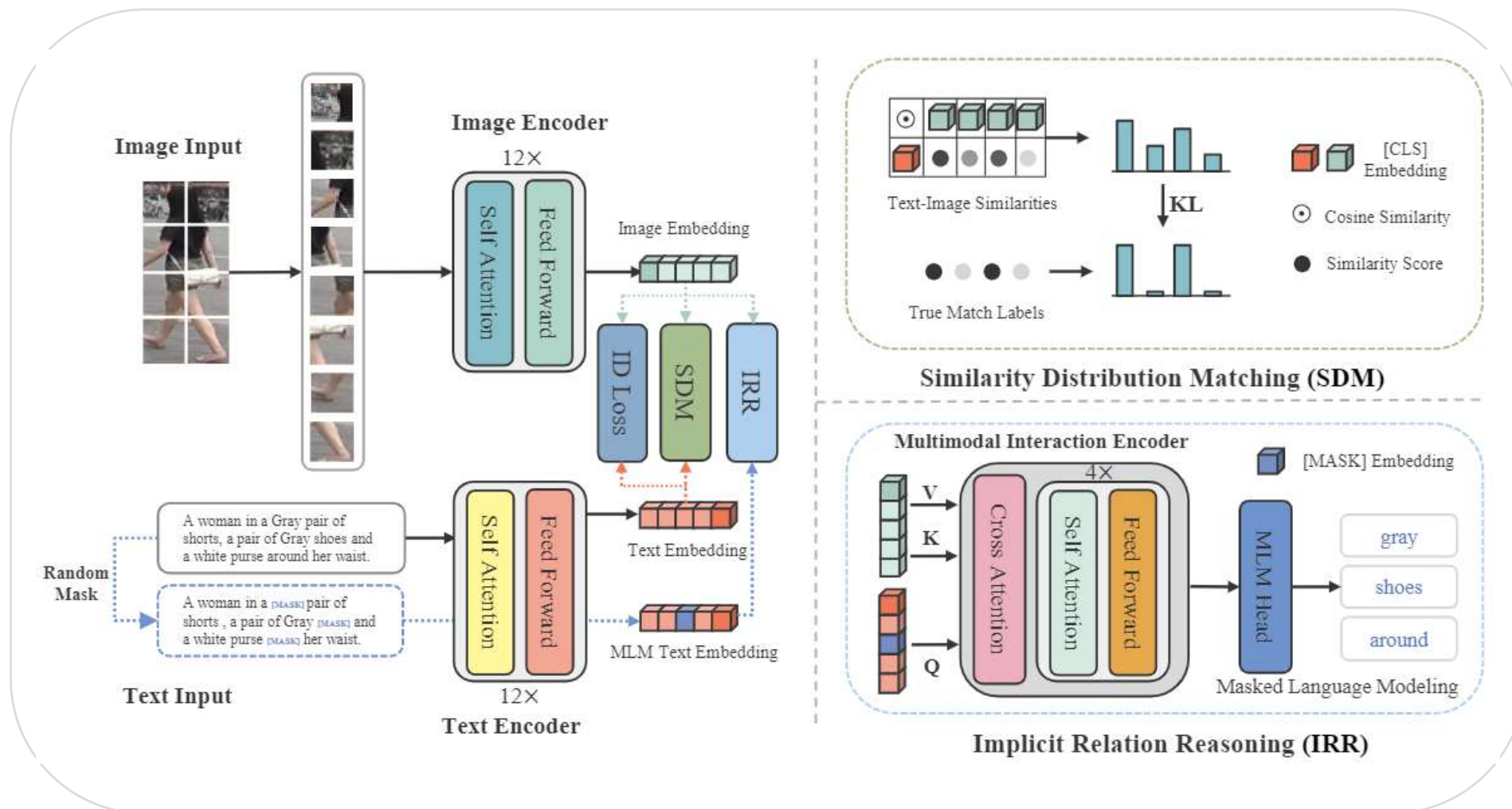
Matches similarity distributions to label distributions

3

FULL CLIP

Fine-tunes both image and text encoders

II. Proposed Method — IRRA Overview



1 • CLIP DUAL ENCODER

2 • SDM + ID

3 • IRR / MLM

Dashed modules: removed at inference

II. Proposed Method — CLIP Dual Encoder

IMAGE ENCODER • ViT-B/16

384×128

patch 16×16

$24 \times 8 = 192$ visual tokens



$[\text{CLS}] \rightarrow f_v$

12 Transformer blocks

TEXT ENCODER • Transformer

BPE vocabulary

49,152

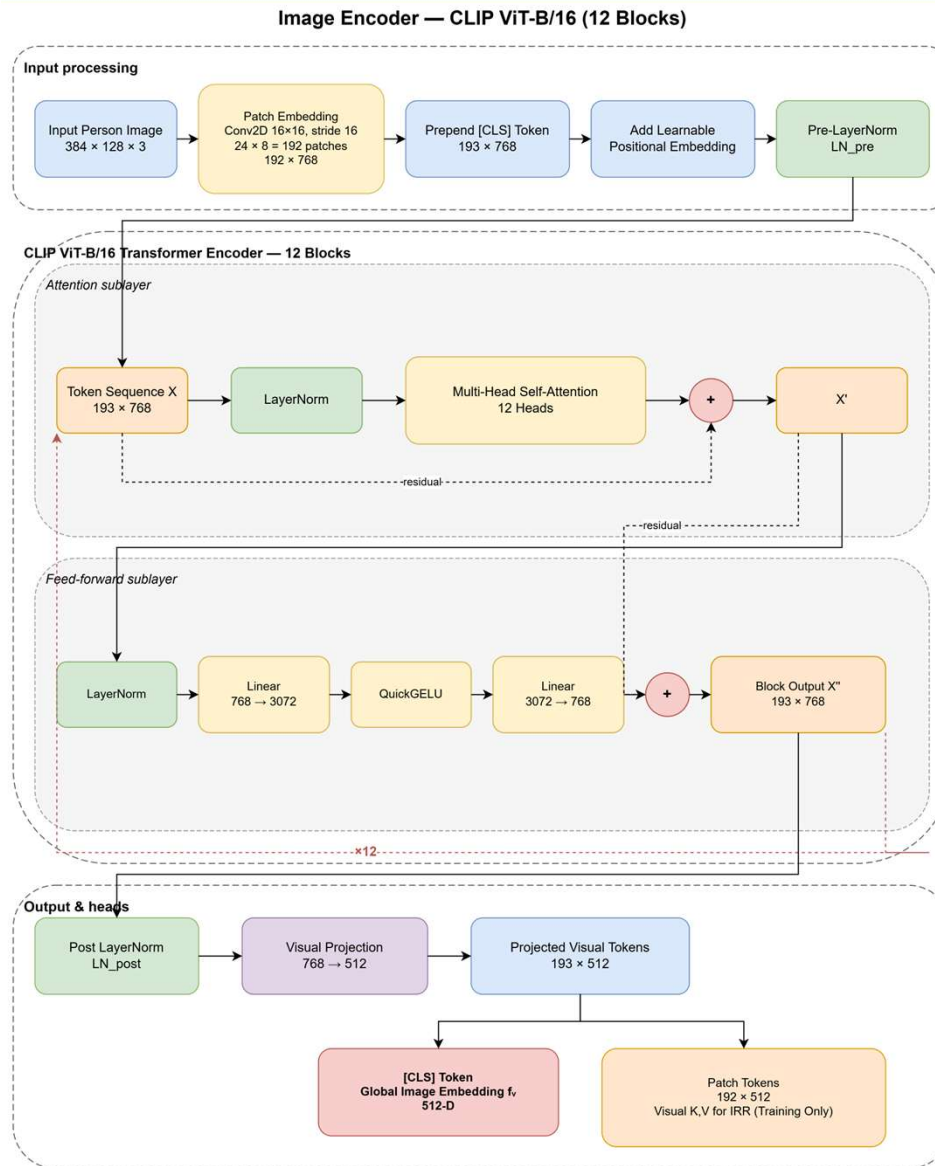
$[\text{SOS}] \text{ token}_1 \dots \text{token}_{77} [\text{EOS}]$



$[\text{EOS}] \rightarrow f_t$

12 Transformer blocks

II. Proposed Method — Inside the ViT Image Encoder



ViT-B/16 • 12 BLOCKS

**1 Patch embedding
+ position**

**2 Pre-LN multi-head
self-attention**

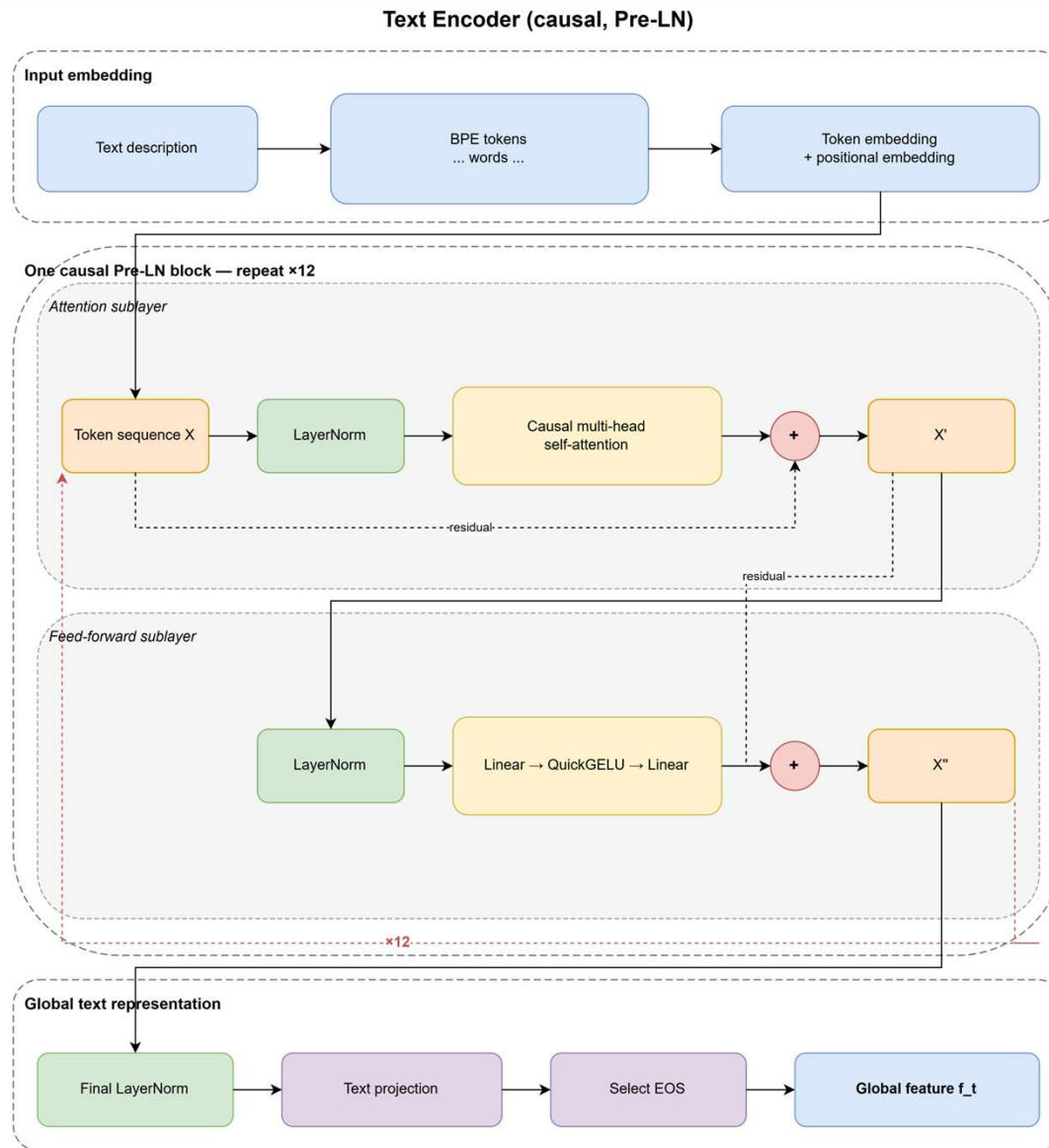
**3 Residual
connection**

**4 Pre-LN MLP: 768
→ 3072 → 768**

**5 [CLS] collects
global visual context**

$$\begin{aligned}x &\leftarrow x + \text{MSA}(\text{LN}(x)) \\x &\leftarrow x + \text{MLP}(\text{LN}(x))\end{aligned}$$

II. Proposed Method — Inside the CLIP Text Encoder



TEXT • 12 BLOCKS

1 BPE token + position

2 Causal self-attention

3 Same Pre-LN residual block

4 [EOS] summarizes the prefix

5 Linear projection → f_t

token i attends only to positions $\leq i$

II. Proposed Method — IRR through MLM

ORIGINAL TEXT

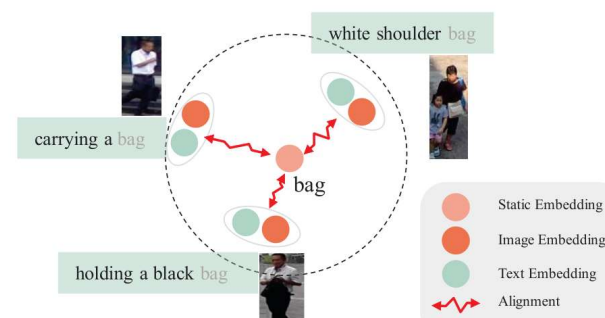
... gray shoes and a white purse around her waist.

MASKED TEXT

... gray [MASK] and a [MASK] purse [MASK] her waist.

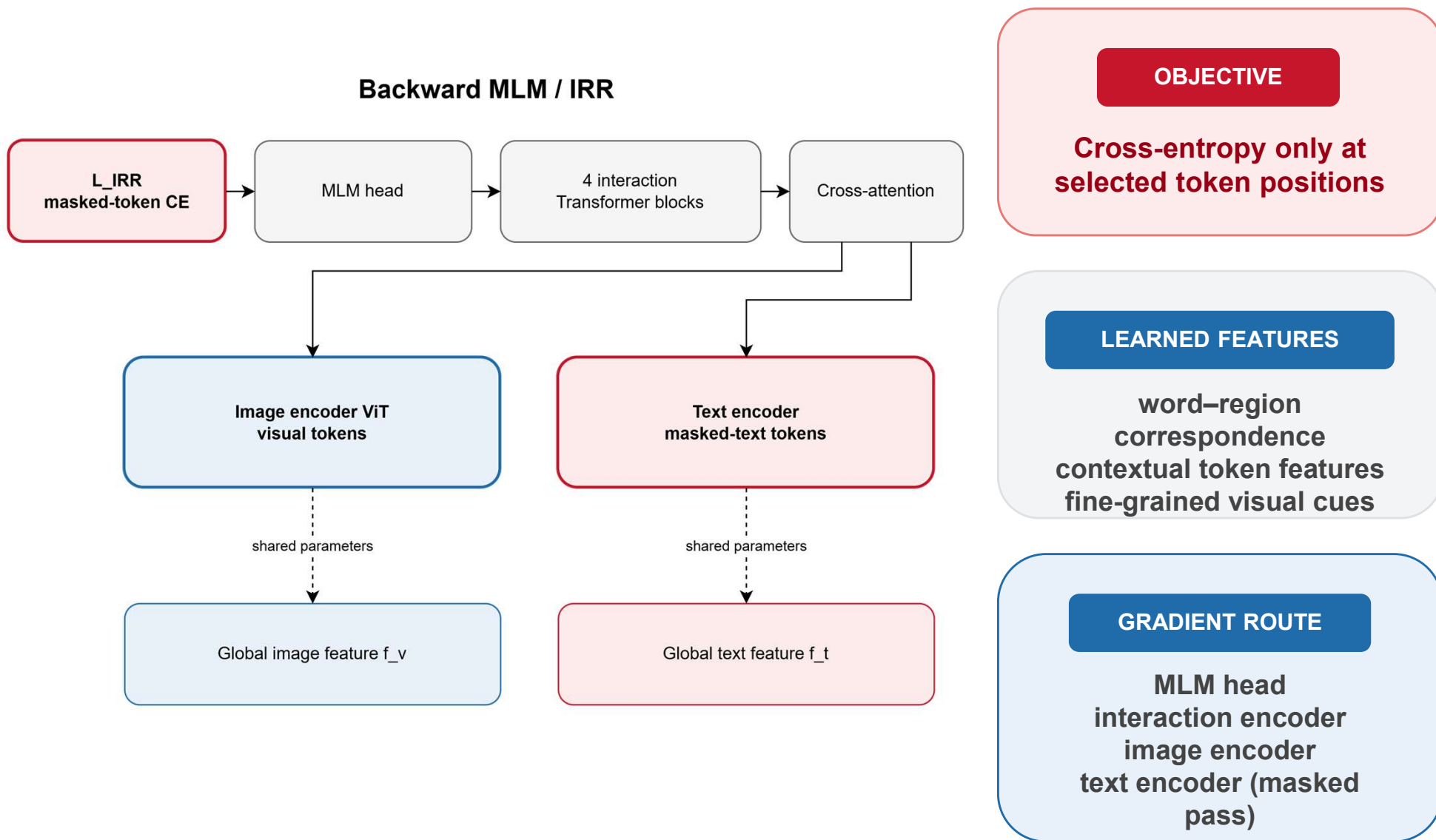
Predictions: shoes • white • around

Select 15% of tokens
80% [MASK] • 10% random • 10%
unchanged

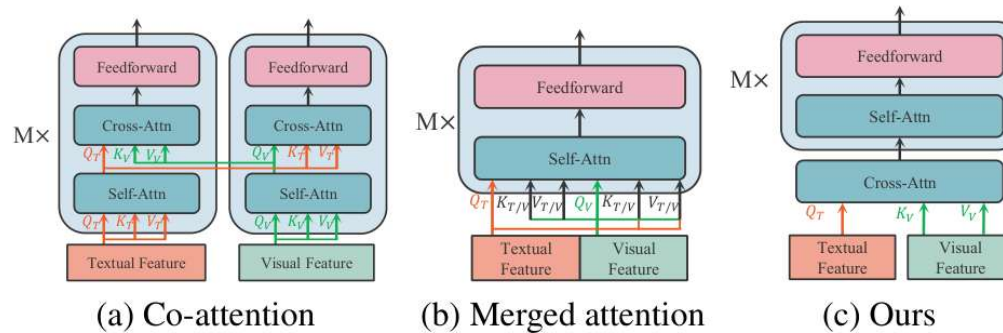


$$\mathcal{L}_{\text{MLM}} = -\frac{1}{|\mathcal{M}|} \sum_{i \in \mathcal{M}} \log p_{\theta}(w_i | \hat{T}, I)$$

II. Proposed Method — What Does the MLM Branch Learn?



II. Proposed Method — Multimodal Interaction



Q masked text tokens

K, V visual tokens

$$\text{MCA}(Q, K, V) = \text{softmax} \left(\frac{QK^T}{\sqrt{d}} \right) V$$

One-way cross-attention $\rightarrow$ 4 Transformer blocks
hidden size = 512 • 8 heads

II. Proposed Method — Similarity Distribution Matching

For a mini-batch of N pairs: $\text{sim}(\cdot, \cdot)$ = cosine similarity; $y_{ij} = 1$ iff the identities match

1 • PREDICTION

$$p^i \rightarrow t_{i,j} = \frac{\exp(\text{sim}(f^v_i, f^t_j)/\tau)}{\sum_{k=1}^N \exp(\text{sim}(f^v_i, f^t_k)/\tau)}$$

2 • TARGET

$$q^i \rightarrow t_{i,j} = \frac{y_{i,j}}{\sum_{k=1}^N y_{i,k}}$$

3 • LOSS

$$\mathcal{L}_{\text{SDM}} = KL(p^i \rightarrow t \parallel q^i \rightarrow t) + KL(p^t \rightarrow i \parallel q^t \rightarrow i)$$

$\tau = 0.02$

Align the full row/column distribution—not only one positive pair.

II. Proposed Method — Identity Classification Loss

$$\mathcal{L}_{\text{iD}} = \frac{1}{2} [CE(Wf^v, y) + CE(Wf^t, y)]$$

INPUT

Global image feature f^v
Global text feature f^t
One shared classifier W

LEARNS

Identity-discriminative
global clusters around
shared class prototypes

BACKPROP

Updates both CLIP
encoders
and classifier W
Classifier removed at
test

**Auxiliary objective: identity classification complements—but does not replace—
cross-modal alignment.**

II. Proposed Method — Training vs. Inference

$$L = L_{IRR} + L_{SDM} + L_{ID}$$

TRAIN

- CLIP dual encoder
- + IRR / interaction encoder
- + MLM head
- + SDM and ID classifier
- → **end-to-end local interaction learning**

INFERENCE

- Only the two encoders remain
- Precomputed gallery embeddings
- One cosine score per image–text pair
- No interaction module required
- → **efficient global retrieval**

Fine-grained learning → efficient global retrieval

III. Experiments — Three Benchmarks

Dataset	Identities	Images	Descriptions	Key property
CUHK-PEDES	13,003	40,206	80,412	2 descriptions/image
ICFG-PEDES	4,102	54,522	54,522	1 description/image
RSTPReid	4,101	20,505	41,010	15 cameras; 5 images/ID

IDENTITY-DISJOINT SPLIT

No identity appears in both training and test sets

CUHK: first benchmark

ICFG: large test gallery

RSTP: multi-camera

Test = generalization to unseen identities

III. Experiments — Metrics and Training Setup

RANK-k

At least one correct image appears in Top-k

mAP

Ranking quality across all relevant images

mINP

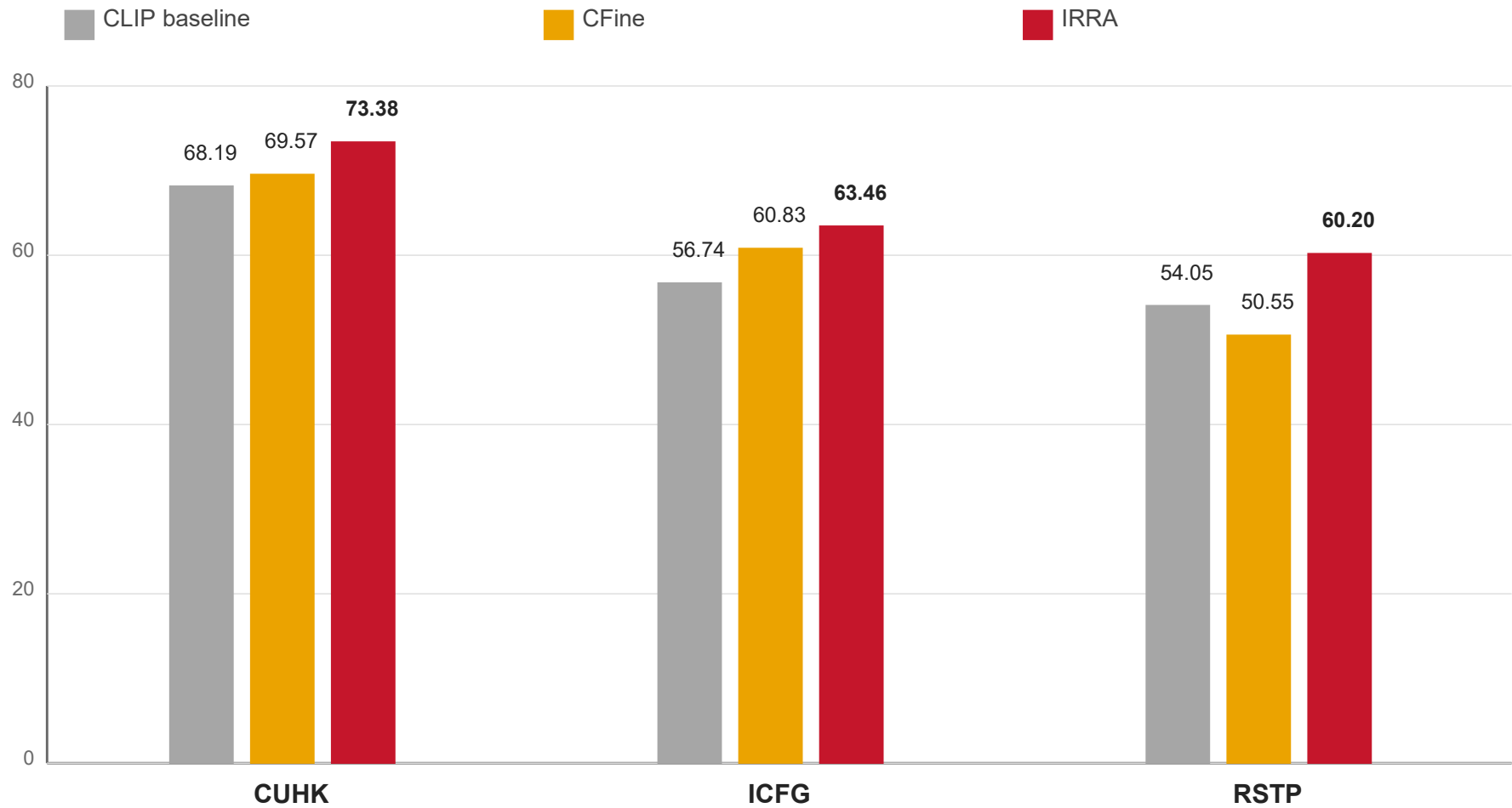
Position of the hardest / last relevant image

IMPLEMENTATION

- CLIP ViT-B/16
- Image size: 384×128
- Text length ≤ 77 tokens
- Adam • 60 epochs
- Learning rate: $1e-5$
- Warm-up: 5 epochs
- New modules: $5e-5$
- $\tau = 0.02$ • RTX 3090

Higher is better for all metrics

III. Results — State-of-the-Art Comparison (2023)

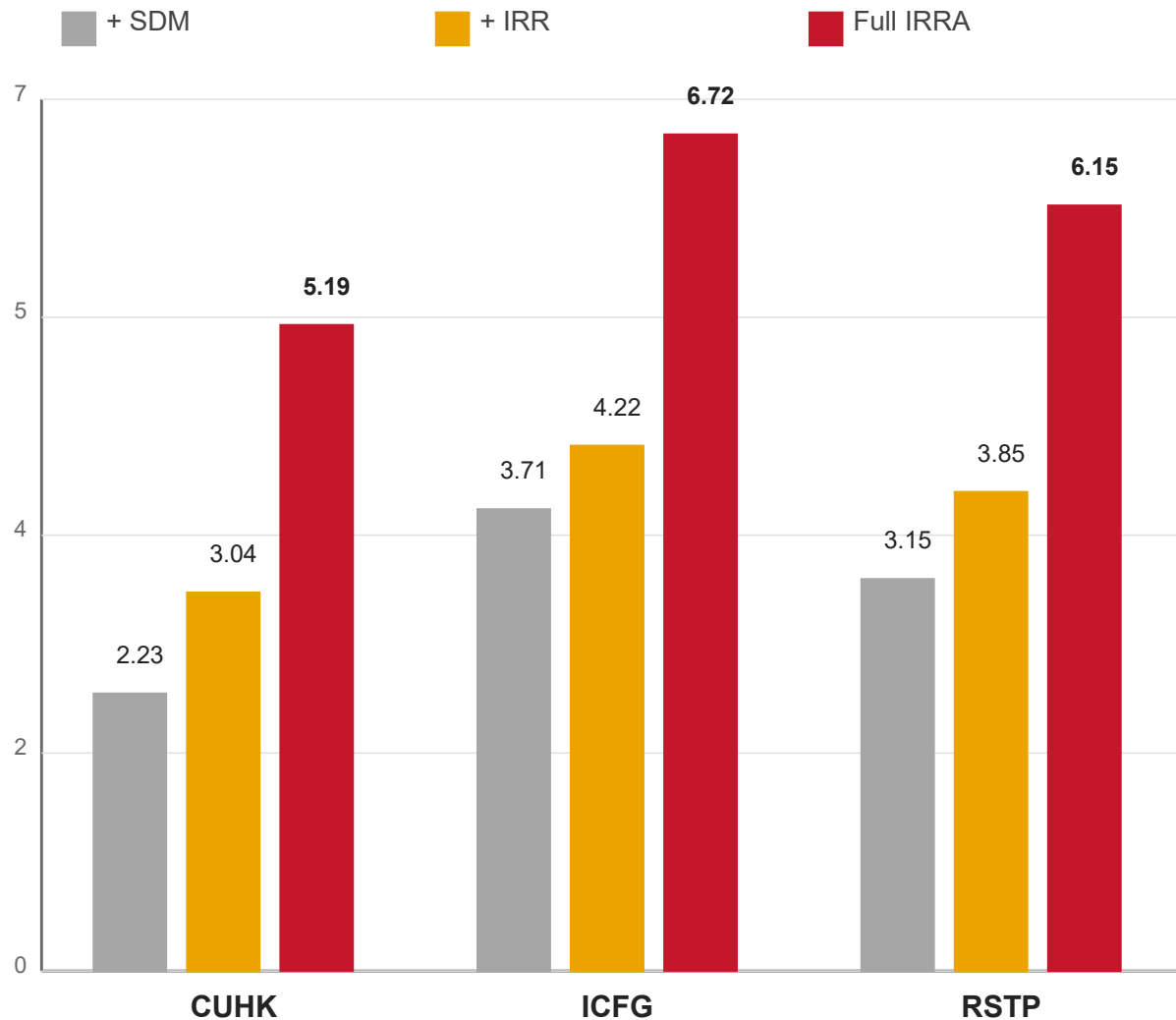


+3.81 vs. CFine

+2.63 vs. CFine

+9.65 vs. CFine

III. Results — Ablation: IRR and SDM



TAKEAWAY

IRR > SDM

R1 gain: +3.04 to +4.22

FULL MODEL WINS

Complementary gains

ID LOSS

Alone, it may reduce R1

CMPM

Weaker than SDM

III. Results — Efficiency and Qualitative Results

INTERACTION MODULE

Module	Params	Time
Co-attn	33.62M	24.30 ms
Merged	12.61M	19.20 ms
Ours	13.66M	6.42 ms

~3× faster than merged attention

Why does IRRA perform better?

IRR exploits subtle discriminative attributes:
clothing • colors • shoes • bags

Interaction module removed at inference

Query: a man crossing the street, wearing a white shirt and black pants, carrying a black backpack

Baseline



IRRA



IV. Discussion — Limitations and Critical Analysis

1 Phrase-level semantics

Token masking may distort phrase-level meaning.

2 Hard positives

ICFG mINP = 7.93: difficult positives remain far down the ranking.

3 Generalization

No cross-dataset, multilingual, or new-domain evaluation.

4 Deployment and ethics

No ANN benchmark; privacy, bias, and surveillance risks.

(1) is acknowledged by the authors; (2)–(4) are additional critical observations.

THREE KEY TAKEAWAYS

1

Local relation learning

during training; global embeddings at inference

2

IRR + SDM

fine-grained interaction and distribution alignment

3

SOTA at CVPR 2023

on CUHK-PEDES, ICFG-PEDES, and RSTPReid

POTENTIAL EXTENSIONS

Phrase masking • hard-sample mining • multilingual retrieval • vector databases

THANK YOU • Q&A

[FULL NAME – STUDENT ID – GROUP]

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Appendix A — SDM vs. InfoNCE and CMPM

Loss	Primary signal	Key observation
InfoNCE	Positive pair contrasted with negatives	The CLIP baseline is already strong
CMPM	Projection matching	Variable scale; distribution sharpness is harder to control
SDM	KL between similarity and label distributions	Temperature control; multiple positives per identity

SDM optimizes each row and column of the similarity matrix—not only one diagonal positive pair.

Appendix B — Complete Results

Dataset	R1	R5	R10	mAP	mINP
CUHK-PEDES	73.38	89.93	93.71	66.13	50.24
ICFG-PEDES	63.46	80.25	85.82	38.06	7.93
RSTPReid	60.20	81.30	88.20	47.17	25.28

Precise wording: “State of the art at publication time, against the baselines evaluated in the paper.”

Appendix C — Frequently Asked Questions

1 Why is it called implicit?

No part labels or local-matching branch at inference.

2 Why does text provide Q?

Masked text tokens query visual tokens to recover the word.

3 Is IRR knowledge lost at inference?

No. Its gradients already updated both encoders end-to-end.

4 Is inference cost really unchanged?

Yes versus the CLIP dual encoder; vector-DB cost was not evaluated.